Fiscal Year:	FY 2022	**Task Last Updated:**	FY 10/28/2021
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Sandra.Olson@nasa.gov	**Fax:**	FY 216 977-7065
PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
PI Address 1:	LTX, Combustion Physics and Reacting Systems Branch		
PI Address 2:	MS 77-5, 21000 Brookpark Rd.		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Changed end date to 12/31/2022, per PI (Ed., 10/3/19) NOTE: Extended to 12/31/2019, per PI (Ed., 1/26/2015) NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David M.S. (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA-STD-6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 pounds per square inch absolute (psia). Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	The centrifuge drop vehicle in the Zero Gravity Research Facility was refurbished in early 2020 to conduct low pressure blowoff testing with 0.318 cm radius polymethyl methacrylate (PMMA) rods to compare with the forced flow data. We completed just a few experiments prior to the pandemic mandatory telework in March 2020, and just began testing again in September 2021. The flame is all blue but is quite stable and does not seem to be strongly affected by Coriolis forces. The results to date of the experiment compare very well with the blowoff branch of the Burning and Suppression of Solids - II (BASS-II) flammability boundary we published in Combustion and Flame in 2017 (Olson S, Ferkul P. "Microgravity flammability boundary for PMMA rods in axial stagnation flow: Experimental results and energy balance analyses." Combustion and Flame. 2017 Jun;180:217-29). The buoyant flow is estimated using Ub = 25 (g/gEarth)^1/2 so the 1g (Earth g) buoyant flow is 25 cm/s for the stagnation geometry. The boundary is just slightly shifted down from the BASS-II boundary. The close alignment of the two boundaries indicates that the square root scaling appears to be valid. The stable flame at lunar g level and 13.8% O2 is very close to the minimum in the BASS-II flammability boundary. This would seem to support prior research that found materials are more flammable on the Moon than on Mars or the Earth. A few additional tests are planned to complete the testing. A paper of the results is planned. During the pandemic, I spent time analyzing flight results from the Solid Inflammatory Boundary at Low Speed (SIBAL) fabrics, since I am burning SIBAL fabric as well as PMMA rods as part of my upcoming experiments on the International Space Station (ISS). I am working on a more traditional flame spread paper for that project, but I did develop a novel analysis of the post-flame smolder wave that developed behind flame, To analyze the smolder wave, I used a variety of techniques to measure the speed of the smolder wave, the number and size of the smolder spots, and the shape of the smolder wave distribution. A paper documenting this analysis is planned.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Abstracts for Journals and Proceedings	Ferkul P, Brown L, Mroczka C, Bhattacharjee S, Fernandez-Pello C, Miller F, Olson S, T'ien J, Wichman I. "Solid Fuel Ignition and Extinction (SoFIE) project on ISS." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Abstracts for Journals and Proceedings	Liao YT, Vetturini AJ, Olson S, Ferkul P. "Concurrent-opposed flame spread reversal phenomenon for ultra-thin solid samples." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Carney A, Li Y, Liao YT, Olson S, Ferkul P. "Concurrent-flow flame spread over thin discrete fuels in microgravity." Combustion and Flame. 2021 Apr;226:211-21. https://doi.org/10.1016/j.combustflame.2020.12.005 , Apr-2021
Articles in Peer-reviewed Journals	Marcum JW, Rachow P, Ferkul PV, Olson SL. "Low pressure flame blowoff of the stagnation region of cast PMMA cylinders in axial mixed convective flow." Combustion and Flame. 2020 Jun;216;385-97. https://doi.org/10.1016/j.combustflame.2020.02.031 , Jun-2020

Articles in Peer-reviewed Journals	Vetturini A, Cui W, Liao YT, Olson S, Ferkul P. "Flame spread over ultra-thin solids: Effect of area density and concurrent-opposed spread reversal phenomenon." Fire Technology. 2020 Jan;56:91-111. https://doi.org/10.1007/s10694-019-00878-w , Jan-2020
NASA Technical Documents	Olson SL, Ferkul PV, Bhattacharjee S, Miller FJ, Fernandez-Pello C, Link S, T'ien JS, Wichman I. "Burning and Suppression of Solids-II (BASS-II) Summary Report." Cleveland, Ohio : NASA Glenn Research Center, 2021. NASA/TM-20210011385. https://ntrs.nasa.gov/search?q=NASA%2FTM-20210011385 , Oct-2021